Relations

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Based on the lecture note uploaded on SPeCTRUM

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Learning Outcome

At the end of this lecture, you should be able to:

☐ Find the cartesian product of two sets.

☐ Understand what is a relation and determine the domain and range of a relation.

☐ Determine if a relation is reflexive.

☐ Determine if a relation is symmetric.

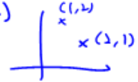
☐ Determine if a relation is transitive.

☐ Determine if a relation is an equivalence relation.

☐ Understand what is a partition and determine the equivalence classes in an equivalent relation.

Cartesian product

An ordered pair is a collection of two objects in a specified order, and is denoted by round bracket, for example (a, b). $(2,1) \neq (1,2)$



for $a \neq b$, $(a,b) \neq (b,a)$
but $\{a,b\} = \{b,a\}$

For two nonempty sets A and B, the Cartesian product (or product set) of A and B, denoted by $A \times B$, is the set of all ordered pairs (a, b) with $a \in A$ and $b \in B$. $(A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\})$

For any two finite, nonempty sets A and B, $n(A \times B) = n(A) \cdot n(B)$.

In general, $A \times B \neq B \times A$.

$A \times A$ often written as A^2.

$|A| = 2$ $\quad$ $|B| = 4$

Example: Let $A = \{a, b\}$ and $B = \{1, 2, 3, 4\}$. Find

$A \times B = \{(a,1), (a,2), (a,3), (a,4), (b,1), (b,2), (b,3), (b,4)\}$

$|A \times B| = 8 = 2 \cdot 4$

$B \times A = \{(1,a), (1,b), (2,a), (2,b), (3,a), (3,b), (4,a), (4,b)\}$

$|B \times A| = 8 = 2 \cdot 4.$

$A^2 = A \times A = \{(a,a), (a,b), (b,a), (b,b)\}$

$|A^2| = 4 = 2 \cdot 2.$

Binary relation

Let A and B be two nonempty sets, a binary relation R from A to B is a subset of $A \times B$.

Let $a \in A$ and $b \in B$, if $(a, b) \in R$, we say that a is R-related to B and denote this by $a\ R\ b$.

If $(a, b) \notin R$, then a is not R-related to B and denote this by $a \not R\ b$.

A binary relation R on A is a subset of $A \times A$.

Example: Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6\}$ and R is a relation from A to B where $(a, b) \in R$ if and only if $b - a = 1$. List the element of R.

$R = \{(1,2), (2,4)\}$

Let $A = \mathbb{Z}$ and a relation R $\underbrace{\text{on } A}$ be defined by: $x\ R\ y \Leftrightarrow x$ divides y. Determine whether the following is true or false.

$A \times A$

$4\ R\ 12$ T	$12\ R\ 4$ F
$3\ R\ 7$ F	$63\ R\ 3$ F

Domain and Range

Let $R \subseteq A \times B$ be a relation from A to B.

The domain of R, $\mathrm{Dom}(R) = \{a \in A \mid a\,R\,b \text{ for some } b \in B\}$
$=$ The set of all the a values of (a, b) in R.

The range of R, $\mathrm{Ran}(R) = \{b \in B \mid a\,R\,b \text{ for some } a \in A\}$
$=$ The set of all the b values of (a, b) in R.

Example: Find the domain and range of the following relations R.

1. $R = \{(a, a), (a, b), (b, b), (b, c), (d, c), (d, e)\}$
   Dom(R) = {a, b, d}
   Ran(R) = {a, b, c, e}
2. $A = \{-2, -1, 0, 1, 2, 3\}$, $B = \{0, 1, 4\}$ and $x\,R\,y \Leftrightarrow y = x^2$.
   relation from A to B.
   R = {(-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4)}
   Dom(R) = {-2, -1, 0, 1, 2}
   Ran(R) = {0, 1, 4}

Matrix representation of relation

Let $A = \{a_1, a_2, \ldots, a_m\}$ and $B = \{b_1, b_2, \ldots, b_n\}$ and let R be a relation from A to B. The matrix of R, denoted by $M_R = [m_{ij}]$ where

$$m_{ij} = \begin{cases} 1 & \text{if } a_i \; R \; b_j; \\ 0 & \text{if } a_i \not{R} \; b_j. \end{cases}$$

Example: Let $A = \{a, b, c\}$, $B = \{1, 2, 4, 5, 6\}$ and let $R = \{(a,2), (a,5), (b,5), (b,6), (c,1), (c,4)\}$ be the relation from A to B.

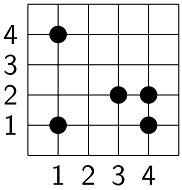
The matrix of $R = M_R =$

$$\begin{array}{c c} & \begin{array}{ccccc} 1 & 2 & 4 & 5 & 6 \end{array} \\ \begin{array}{c} a \\ b \\ c \end{array} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

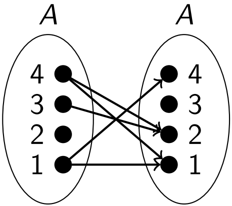
Other representations of relation

Let $A = \{1, 2, 3, 4\}$ and $R = \{(1,1), (1,4), (3,2), (4,1), (4,2)\}$ be the relation on A.

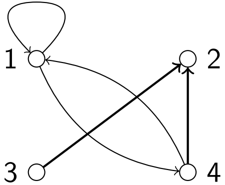
1. Coordinate diagram



2. Arrow diagram



3. Digraph (only for A to A)



Properties of relations

Let R be a relation on a set A.

Reflexive

R is reflexive if $a \mathrel{R} a, \forall a \in A$. (Every element $a \in A$ is related to itself.)

R is not reflexive if $\exists a \in A$, such that $(a, a) \notin R$.

Symmetric

R is symmetric if $\forall a, b \in A, \left(a \mathrel{R} b \to b \mathrel{R} a.\right)$

R is not symmetric if $\exists (a, b) \in R$, but $(b, a) \notin R$.

Transitive

R is transitive if $\forall a, b, c \in A, \left(a \mathrel{R} b \text{ and } b \mathrel{R} c \to a \mathrel{R} c.\right)$

R is not transitive if $\exists (a, b) \in R$ and $(b, c) \in R$, but $(a, c) \notin R$. If there is no such pairs of (a, b) and (b, c) in R, then R is considered transitive.

Properties of relations

Example: Let $A = \{1, 2, 3, 4, 5\}$.
Is each of the following relations (i) reflexive (ii) symmetric (iii) transitive?

1. $R_1 = \{(1,1), (1,2), (2,1), (2,2), (2,4), (3,3), (4,2), (4,4), (5,5)\}$.
   R_1 is reflexive and symmetric.
   R_1 is not transitive because $(1,2) \in R_1$, $(2,4) \in R_1$ but $(1,4) \notin R_1$
2. $R_2 = \{(1,1), (1,3), (2,1), (2,3), (3,1), (3,3), (5,5)\}$.
   R_2 is not reflexive because $2 \in A$ but $(2,2) \notin R_2$
   R_2 is not symmetric because $(2,1) \in R_2$ but $(1,2) \notin R_2$
   R_2 is transitive because for all $a,b,c \in A$, if aR_2b and bR_2c, then aR_2c.
3. $R_3 = \{(2,2), (3,3)\}$.
   R_3 is not reflexive, because $1 \in A$ but $(1,1) \notin R_3$
   R_3 is symmetric and transitive.
4. $R_4 = \{(a,b) \in A \times A \mid a \geq b\}$. $\forall a \in A$, $a \geq a$, so aR_4a and R_4 is reflexive.

Since $(2,1) \in R_4$ but $(1,2) \notin R_4$ so R_4 is not symmetric.
$\forall a,b,c \in A$, if $(a,b) \in R_4$ and $(b,c) \in R_4$, then $a \geq b$ and $b \geq c$. So $a \geq b \geq c$ and $(a,c) \in R_4$. So R_4 is transitive.

A relation R on a set A is called an equivalence relation if R is reflexive, symmetric and transitive.

Example: Let S be the relation on the set of all real numbers defined by $S = \{(a, b) \in \mathbb{R} \times \mathbb{R} \mid a = b\}$. Prove that S is an equivalence relation on $\mathbb{R}$.

domain

Reflexive: Let $a \in \mathbb{R}$. Then it is clear that $a = a$. So, $(a, a) \in S$ and S is reflexive.

Symmetric: Let $a, b \in \mathbb{R}$ and $(a, b) \in S$. Then $a = b$, and so $b = a$. So, $(b, a) \in S$ and S is symmetric.

Transitive: Let $a, b, c \in \mathbb{R}$ and $(a, b) \in S$, $(b, c) \in S$. Then $a = b$ and $b = c$, and so $a = b = c$. So, $(a, c) \in S$ and S is transitive.

Since S is reflexive, symmetric and transitive, S is an equivalence relation on $\mathbb{R}$.

Partition

Let A be a set and let $A_1, A_2, \ldots, A_n \subseteq A$ such that

(i) $A_1 \cup A_2 \cup \cdots \cup A_n = A$ and

(ii) $A_i \cap A_j = \emptyset$ for $A_i \neq A_j$,

then $A_1, A_2, \ldots, A_n$ is called a partition of A.

$A_1, \ldots, A_n$ are not intersecting with each other.

Example: Let $A = \{a, b, c, d, e, f, g\}$. Which of the following is/are partition(s) of A?

1. $A_1 = \{a, c\}$, $A_2 = \{b, g\}$, $A_3 = \{d, e, f\}$

$A_1 \cup A_2 \cup A_3 = \{a, b, c, d, e, f, g\}$ and $A_1 \cap A_2 = \phi$, $A_1 \cap A_3 = \phi$, $A_2 \cap A_3 = \phi$.

So A_1, A_2, A_3 is a partition of A.

2. $B_1 = \{a, g\}$, $B_2 = \{c\}$, $B_3 = \{d, g\}$, $B_4 = \{b, e, f\}$

Notice that $B_1 \cap B_3 = \{g\} \neq \phi$.

So, B_1, B_2, B_3, B_4 is not a partition of A.

3. $C_1 = \{c\}$, $C_2 = \{a, b, g\}$, $C_3 = \{d, f\}$

Notice that $C_1 \cup C_2 \cup C_3 = \{a, b, c, d, f, g\} \neq A$

So, C_1, C_2, C_3 is not a partition of A.

Partition

Example: For each $i = 0, 1, 2, 3, 4$, let $A_i = \{n \in \mathbb{Z} \mid n = i + 5k, \text{ for some } k \in \mathbb{Z}\}$. Is A_0, A_1, A_2, A_3, A_4 a partition of $\mathbb{Z}$?

$A_0 = \{n \in \mathbb{Z} \mid n = 5k \text{ for some } k \in \mathbb{Z}\} = \{\ldots, -15, -10, -5, 0, 5, 10, 15, \ldots\}$

$A_1 = \{n \in \mathbb{Z} \mid n = 5k+1 \text{ for some } k \in \mathbb{Z}\} = \{\ldots, -14, -9, -4, 1, 6, 11, 16, \ldots\}$

$A_2 = \{n \in \mathbb{Z} \mid n = 5k+2 \text{ for some } k \in \mathbb{Z}\} = \{\ldots, -13, -8, -3, 2, 7, 12, 17, \ldots\}$

$A_3 = \{n \in \mathbb{Z} \mid n = 5k+3 \text{ for some } k \in \mathbb{Z}\} = \{\ldots, -12, -7, -2, 3, 8, 13, 18, \ldots\}$

$A_4 = \{n \in \mathbb{Z} \mid n = 5k+4 \text{ for some } k \in \mathbb{Z}\} = \{\ldots, -11, -6, -1, 4, 9, 14, 19, \ldots\}$

Notice that $A_0 \cup A_1 \cup A_2 \cup A_3 \cup A_4 = \mathbb{Z}$ and $A_i \cap A_j = \phi$ for $A_i \neq A_j$, $i, j = 0, 1, 2, 3, 4$. Therefore, A_0, A_1, A_2, A_3, A_4 is a partition of $\mathbb{Z}$.

Equivalence class

Let R be an equivalence relation on a set A. For any element $a \in A$, the equivalence class of a with respect to R, denoted by $[a]$ (or $[a]_R$), is the set

$$[a] = \{b \in A \mid (a, b) \in R\}.$$

Example: Let $A = \{1, 2, 3, 4, 5, 6\}$ and let $R = \{(1, 1), (1, 5), (2, 2), (2, 3), (2, 6), (3, 2), (3, 3), (3, 6), (4, 4), (5, 1), (5, 5), (6, 2), (6, 3), (6, 6)\}$.
Find all the equivalence class in R.

$[1] = \{1, 5\} = [5]$ $\qquad$ $[4] = \{4\}$

$[2] = \{2, 3, 6\} = [3] = [6]$.

So, $\{1,5\}$, $\{2,3,6\}$, $\{4\}$ are 3 distinct equivalence classes in R.

Notice that $[1] \cup [2] \cup [4] = A$ and $A_i \cap A_j = \emptyset$ $\forall i = 1, 2, 4, i \neq j$.

Therefore $[1], [2], [4]$ is a partition of A.

We call $[1], [2], [4]$ is a partition of A induced by R.

Equivalence class

Let $m, n \in \mathbb{Z}$ and $d \in \mathbb{Z}^+$. Then, m is congruent to n modulo d (denoted by $m \equiv n \pmod{d}$) means that $d|(m-n)$.

All of the followings are equivalence:

- $m \equiv n \pmod{d}$. $\Leftrightarrow m = kd + n$ for some $k \in \mathbb{Z}$.
- $d|(m-n)$.
- $\frac{m-n}{d} = k$ for some $k \in \mathbb{Z}$.
- $m - n = kd$ for some $k \in \mathbb{Z}$.
- $m = kd + n$ for some $k \in \mathbb{Z}$.

Example:

$13 \equiv 3 \pmod{5} \equiv 8 \pmod{5} \equiv -2 \pmod{5}$

$\therefore 5|(13-3)$ and $5|(13-8)$ and $5|(13-(-2))$ or

$13 = 2(5) + 3$ and $13 = 1(5) + 8$ and $13 = 3(5) + (-2)$.

integer k integer k integer k.

Let R be the relation on the set of all positive integers defined by $R = \{(x,y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ \mid x \equiv y \pmod 3\}$. Show that R is an equivalence relation. Then, partition $\mathbb{Z}^+$ into equivalence classes.

integer k

(R) Let $a \in \mathbb{Z}^+$. We can always write $a = 3(0) + a$. So, $a \equiv a \pmod 3$ and $(a,a) \in R$. Therefore, R is reflexive.

(S). Let $a, b \in \mathbb{Z}^+$ and $(a,b) \in R$. Then $a \equiv b \pmod 3$ and $a = 3k + b$ for some integer k. Then $b = a - 3k = 3(-k) + a$ where $-k \in \mathbb{Z}$. So $b \equiv a \pmod 3$ and $(b,a) \in R$. Therefore R is symmetric.

(T) Let $a, b, c \in \mathbb{Z}^+$ and $(a,b) \in R, (b,c) \in R$. Then, $a \equiv b \pmod 3$ and $b \equiv c \pmod 3$. Then, $a = 3k_1 + b$ and $b = 3k_2 + c$ for some integers k_1 and k_2. Then, $a = 3k_1 + (3k_2 + c) = 3(k_1 + k_2) + c$ where $k_1 + k_2 \in \mathbb{Z}$. So, $a \equiv c \pmod 3$ and $(a,c) \in R$. So R is transitive.

Since R is reflexive, symmetric and transitive, R is an equiv. relation on $\mathbb{Z}^+$

Equivalence class

$[1] = \{1, 4, 7, 10, \dots\} = [4] = [7] = \dots = [3k+1] = \dots$

$[2] = \{2, 5, 8, 11, \dots\} = [5] = [8] = \dots = [3k+2] = \dots$

$[3] = \{3, 6, 9, 12, \dots\} = [6] = [9] = \dots = [3k] = \dots$

where $k = 0, 1, 2, \dots$

Notice that $[1] \cup [2] \cup [3] = \mathbb{Z}^+$ and $[i] \cap [j] = \emptyset$ for $i, j = 1, 2, 3$, $i \neq j$.

Therefore $[1], [2], [3]$ is a partition of $\mathbb{Z}^+$ induced by R; and $[1], [2], [3]$ are equivalence classes of $\mathbb{Z}^+$.